Android SDK

Ver: 2.3.8

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About:

This document provides the functional and implementation information to work with our Iris Scanner. By using these SDK functions, you can get camera image preview, enroll iris template, capture kinds of IREX image(K1/K2/K3/K7 image), and identify users by templates. This SDK provided facility to extract different iris formats like

1.Raw and BMP for background image
2.Raw and BMP for K1,K2,K3,K7 image
3.Jp2 data
4.ISO iris data

Dependency:

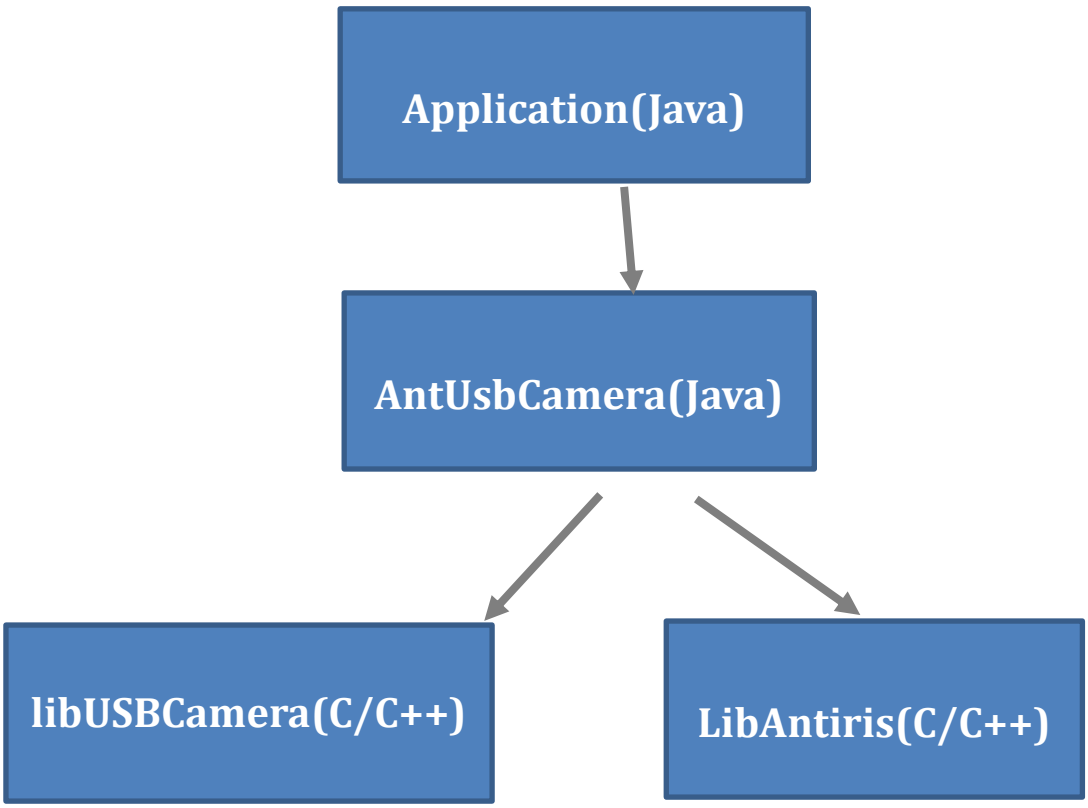
1. USB HOST ENABLED in device

2. Android SDK version: 31

3. Develop tools: Android Studio

4.Library:
antiris.antusbcamera.jar
lib_Ant_iris.so(iris algorithm lib)
libUSBCamera.so(usb camera control lib)
libopenjpeg.so(jpeg encoder lib)

Module Relationship Diagram



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getInstance		
Get unique USBControl instance		
return	**Type**	**Detail**
	static USBControl	

init		
Initialize Camera connection		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
activity	Activity	Main Activities context
cameraView	View	Camera image view
listener	DevConnectListener	Device connect listener context onAttachDev: found a valid usb camera onDettachDev: usb camera unplugged onConnectDev: usb camera has been opened onPreview: started or stoped

		preview
Start preview	boolean	Start preview after open device

isCameraOpened		
Judge whether the camera is on/off		
return	**Type**	**Detail**
	boolean	

antRequestPermission		
Request USB permission		
return	**Type**	**Detail**
	null	

antIrisEnroll		
Start enrolling iris		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
username	String	User Name of template
temple	byte[]	Default null, if not be null, selected template will be rewritten
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing
timeout	long	duration of enroll (ms)

enrollCallback	EnrollCallback	callback of enroll result onSuccess() onError onProcess

antStopEnroll		
Stop enrolling iris		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
showMsg	boolean	Feedback error

antIrisIrexCapture		
Start irex capture iris		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
username	String	User Name of template
IREX Capture Type	int	1;2;3;7 default 7
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing
manual	boolean	Manual capture
timeout	long	Duration of capture (ms)
irexCaptureCallback	IrexCaptureCallback	callback of capture result onSuccess() onError onProcess

antStopIrexCapture		
Stop irex capture iris		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
showMsg	boolean	Feedback error

antIrisIdentify		
Start identifying iris		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing
VerifyCallback	VerifyCallback	callback of identify result onSuccess() onError onProcess
templates	Byte[]	One or more templates , size: 6820 *N
count	int	number of templates

antStopIdentify

Stop identifying iris		
return	**Type**	**Detail**
	null	

isEnrolling		
Judge whether enrolling iris		
return	**Type**	**Detail**
	boolean	

isIdentifying		
Judge whether Identifying iris		
return	**Type**	**Detail**
	boolean	

isPreviewing		
Judge whether is previewing		
return	**Type**	**Detail**
	boolean	

stopStream		
Stop or start camera stream,depending on isPreviewing		
return	**Type**	**Detail**
	null	

Param	Type	Detail

release		
Release usb camera objects		
return	**Type**	**Detail**
	null	

antSetExposure		
Set exposure of camera		
return	**Type**	**Detail**
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing
exposure	int	0 ~ 1000

antSetEyeNum		
set eye number before enroll.		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing

getPreviewWidth		
Get preview image width		
return	**Type**	**Detail**
width	int	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

getPreviewHeight		
Get preview image width		
return	**Type**	**Detail**
height	int	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

getBackgroundBytes		
Get realtime raw data of camera,		
return	**Type**	**Detail**
realtime raw data	byte[]	size = getPreviewWidth * getPreviewHeight , can be saved to file
Param	**Type**	**Detail**

flip	boolean	whether to flip horizontally

getBackgroundBitmap		
Get realtime bmp data of camera,		
return	**Type**	**Detail**
bmp data	byte[]	can be saved to file
Param	**Type**	**Detail**
flip	boolean	whether to flip horizontally

getCameraSerialNumber		
Get USB camera serial number		
return	**Type**	**Detail**
serial number	String	

indicatorLEDOnOff		
Turn on/off Indicator light		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing
on	boolean	

IRLED_OnOff		
Turn on/off LED light. Only used in some one model		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
on	boolean	

antSetStatusLed		
Set LED status		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
Invalue	Int	0 off 1 Red 2 Green 3 Red+Green 4 Red flash rapidly 5 Red flash slowly 6 Green flash rapidly 7 Green flash slowly 8 Red+Green flash slowly

getK7Image		
Get K7 raw image		
return	**Type**	**Detail**
raw data	byte[]	size = getK7ImgWidth * getK7ImgHeight , can be saved to file
Param	**Type**	**Detail**
eyeNum	int	Eye index:

		0 for single left eye processing 1 for single right eye processing

getK7Bitmap		
Get K7 bmp image		
return	**Type**	**Detail**
bmp data	byte[]	can be saved to file
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

getK7J2kData		
Get j2k data . It is encoded by JPEG 2000		
return	**Type**	**Detail**
Jp2 data	byte[]	can be saved to file
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

getK7ImgWidth		
Get K7 raw image width		
return	**Type**	**Detail**
width	int	

Param	Type	Detail
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

getK7ImgHeight		
Get K7 raw image height		
return	**Type**	**Detail**
height	int	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

getk7Quality		
Get k7 image quality.		
return	**Type**	**Detail**
quality	int	
Param	**Type**	**Detail**
eyeNum	int	Eye index: 0 for single left eye processing 1 for single right eye processing

EnrollCallback. onSuccess IrexCaptureCallback. onSuccess

Callback for enroll success

return	Type	Detail
	null	
Param	**Type**	**Detail**
Eye number	Int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing
User name	String	User name
template	Byte []	Template after enroll success

EnrollCallback. onError **IrexCaptureCallback. onError**
Callback for enroll error

return	Type	Detail
	null	
Param	**Type**	**Detail**
msg	String	Error msg string

EnrollCallback. onProcess **IrexCaptureCallback. onProcess**
Callback for enroll progress

return	Type	Detail
	null	
Param	**Type**	**Detail**
Eye number	Int	Eye index: 0 for single left eye processing 1 for single right eye processing 2 for both eyes processing

progress	Int	Enroll progress ,0 - 100
distance	Int	Distance from eyes to sensor.
opening	Int	Degree of eyes opening

IdentifyCallback. onSuccess		
Callback for identify success		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
distance	Int	Distance from eyes to sensor
User name	String	User name

IdentifyCallback. onError		
Callback for identify error		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
msg	String	Error msg string

IdentifyCallback. onProcess		
Callback for identify progress		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
distance	Int	Distance from eyes to sensor.
opening	Int	Degree of eyes opening

saveGrayToPng		
Save gray data to png file (use 'FileUtil.makeDir' to create directory for the first)		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
data	Byte[]	Gray data(8bit)
FilePath	String	Full File path
width	int	Image width
height	int	Image height

saveGrayToBmp		
Save gray data to bmp file.		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
data	Byte[]	Gray data(8bit)
dir	String	File directory
fileName	String	File name
width	int	Image width
height	int	Image height
flip	Boolean	Horizontal flip

SaveGrayToJp2		
Save gray data to jp2 file		
return	**Type**	**Detail**
	null	

Param	Type	Detail
data	Byte[]	Gray data(8bit)
dir	String	File directory
fileName	String	File name
width	int	Image width
height	int	Image height

grayToJp2		
Gray data to jp2 data		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
data	Byte[]	Gray data(8bit)
width	int	Image width
height	int	Image height
Jp2 data	Byte[]	Jp2 data(8bit)
Out size	int	jp2 data size
Image type	int	1;2;3;7; Default to 7
Compress mode	int	PREVIOUS COMPRESSION UNDEF=0 PREVIOUS COMPRESSION LOSSLESS OR NONE=1 PREVIOUS COMPRESSION LOSSY=2 Default to 2

antIrisSetEnrollIREXType		
Set enroll irex type		
return	**Type**	**Detail**
	null	
Param	**Type**	**Detail**
type	int	1;2;3;7

isMonocular		
Determine if it is a single eye		
return	**Type**	**Detail**
	boolean	

IrisIso. generate		
Generate .iso file		
return	**Type**	**Detail**
data	Byte[]	Iso data
Param	**Type**	**Detail**
version	int	Defult:0
data	Byte[]	Jp2 image data
width	int	Jp2 image width
height	int	Jp2 image height
uid	Byte[]	camera id
quality	Int	K Image quality (0 – 100)
Image type	Int	K image type (1,2,3,7) Default to 7
Image Format	Int	Default to 10(JPEG2000)
Image properties bit field	int	PREVIOUS COMPRESSION UNDEF=0 PREVIOUS COMPRESSION LOSSLESS OR NONE=1 PREVIOUS COMPRESSION LOSSY=2

		Default to 2
Bit depth	Int	Default to 8